Dynamic Data Masking

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Dept: Data Engineering

Task: Implement dynamic data masking on specific columns (e.g., email addresses or credit card numbers) in a Synapse table to restrict data visibility based on user roles.

In synapse, dedicated sql pool create table with masked columns

Synapse live | Validate all | Publish all 4

Develop

Filter resources by name

SQL scripts 4

- SQL script 1
- SQL script 2
- SQL script 3
- SQL script 4

SQL script 1 | SQL script 2 | SQL script 3 | SQL script 4

Other users in your workspace may have access to modify this item. Do not use this may have access to the workspace.

Run | Undo | Publish | Query plan | Connect to sqlpooldemo01 | Use database sqlpooldemo01

Dedicated SQL pool, sqlpooldemo01, is paused. Go to Manage pools to resume your pool.

```sql
CREATE TABLE employees (
    id INT,
    first_name VARCHAR(255),
    last_name VARCHAR(255),

    -- Sensitive columns with masking
    salary INT MASKED WITH (FUNCTION = 'default()'),
    phone VARCHAR(15) MASKED WITH (FUNCTION = 'partial(0,"XXXXXX",4)'),
    email VARCHAR(255) MASKED WITH (FUNCTION = 'email()')
);

-- Insert sample data into the employees table
INSERT INTO employees VALUES (1, 'Aakash', 'Sharma', 75000, '9876543210', 'aakash@gmail.com')
INSERT INTO employees VALUES (2, 'Riya', 'Patel', 82000, '9123456780', 'riya.patel@gmail.com')
INSERT INTO employees VALUES (3, 'Karan', 'Mehta', 68000, '9988776655', 'karan.mehta@gmail.com')


SELECT * FROM employees;
```

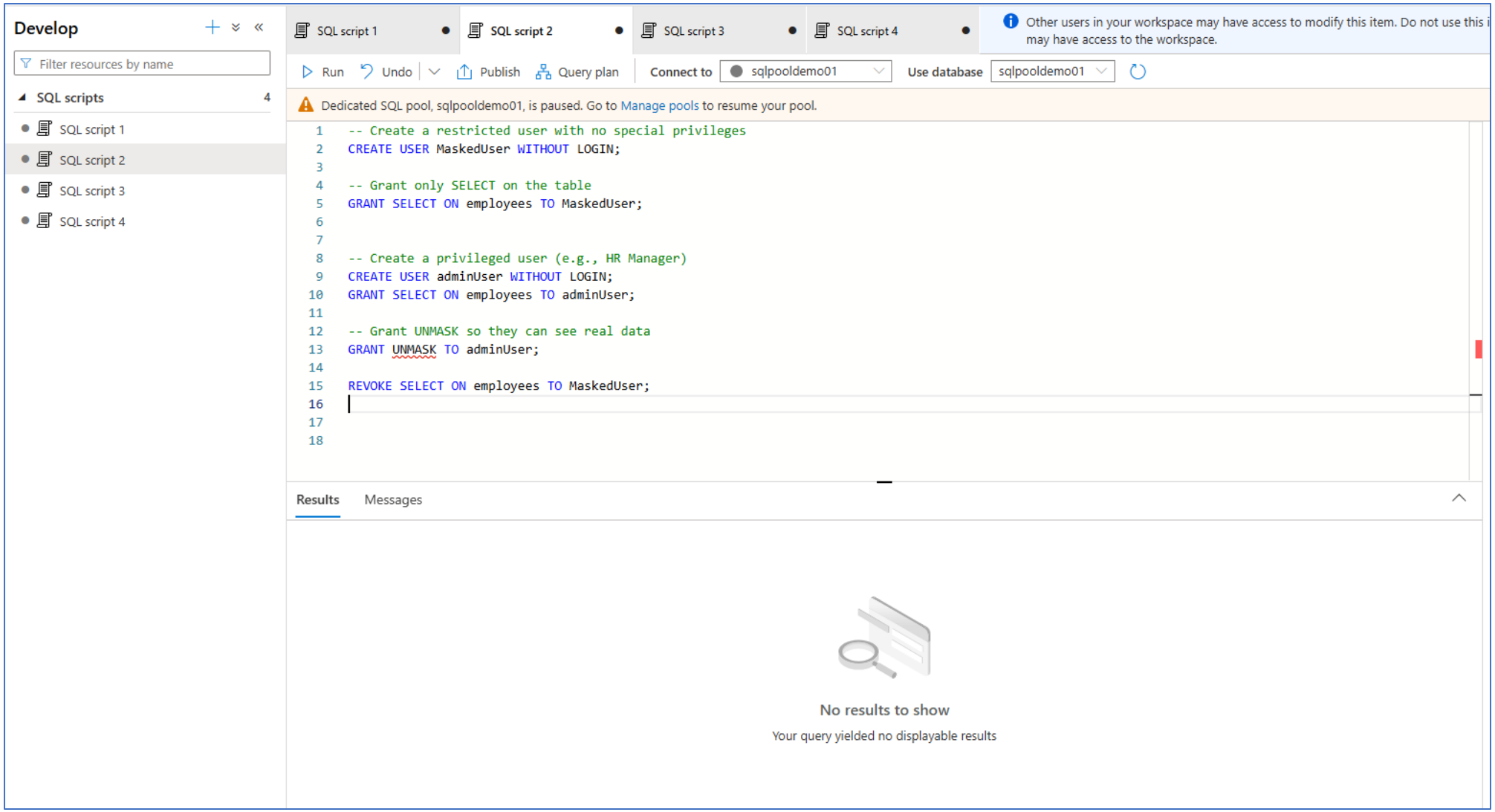
Results | Messages

View: Table | Chart | Export results

Search

id	first_name	last_name	salary	phone	email
3	Karan	Mehta	68000	9988776655	karan.mehta@...
2	Riya	Patel	82000	9123456780	riya.patel@gma...
1	Aakash	Sharma	75000	9876543210	aakash@gmail....

Create specific users and grant unmask to some of them



Execute the query for normal user (to see masked colum values

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Run | Undo | Publish | Query plan | Connect to sqlpooldemo01 | Use database sqlpooldemo01

Dedicated SQL pool, sqlpooldemo01, is paused. Go to Manage pools to resume your pool.

```sql
-- Execute SELECT as the masked user
EXECUTE AS USER = 'MaskedUser';

SELECT * FROM employees;

-- Revert back to admin
REVERT;


-- Execute as admin User
EXECUTE AS USER = 'adminUser'

SELECT * FROM employees;

REVERT;
```

Results | Messages

View: Table | Chart | Export results

Search

id	first_name	last_name	salary	phone	email
3	Karan	Mehta	1004	XXXXXX6655	kXXX@XXXX.co...
2	Riya	Patel	1990	XXXXXX6780	rXXX@XXXX.com
1	Aakash	Sharma	1665	XXXXXX3210	aXXX@XXXX.co...

AdminUser can still access all details

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REVERT;
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Results | Messages

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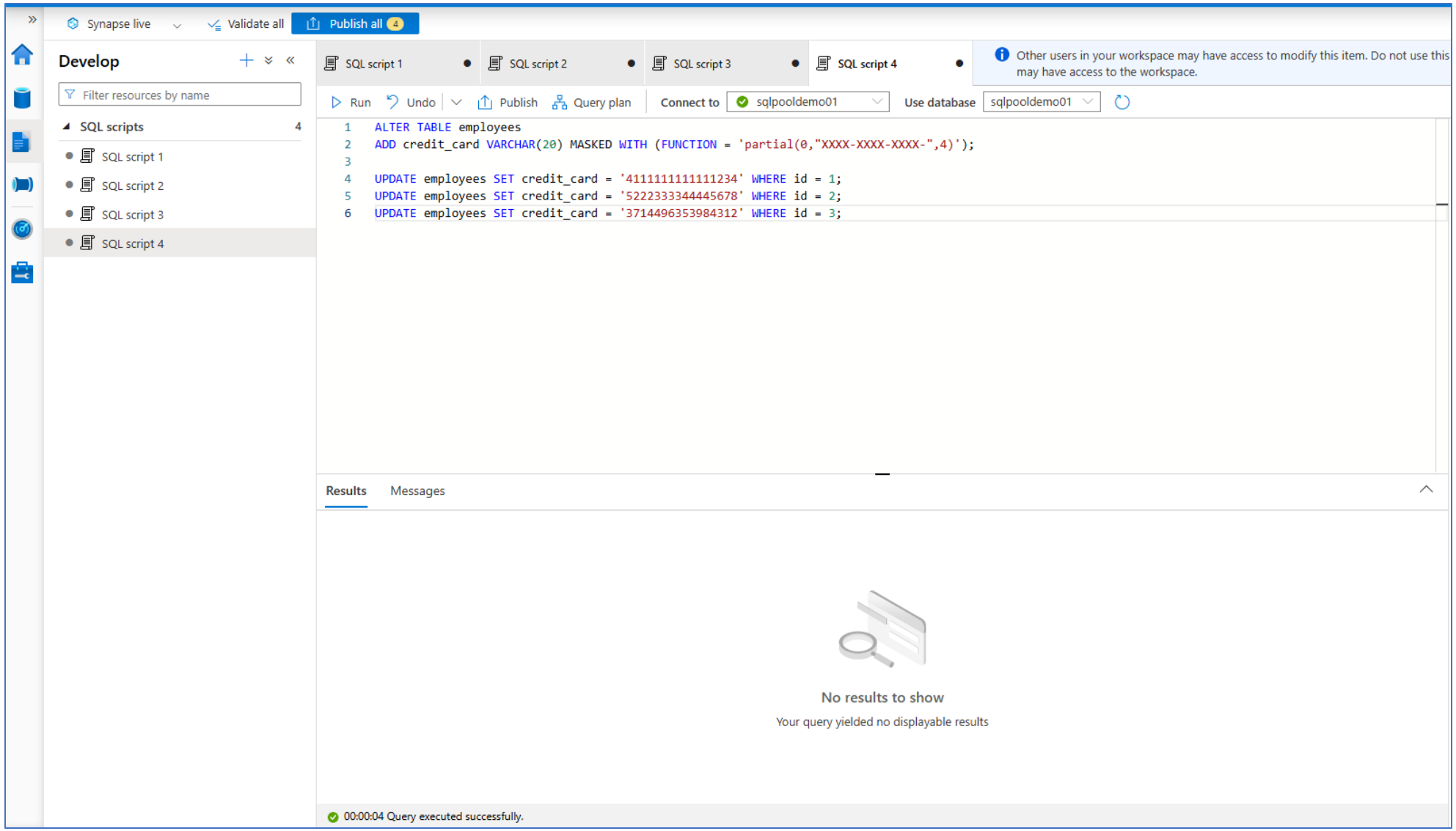
Search

id	first_name	last_name	salary	phone	email
3	Karan	Mehta	68000	9988776655	karan.mehta@...
2	Riya	Patel	82000	9123456780	riya.patel@gma...
1	Aakash	Sharma	75000	9876543210	aakash@gmail....

Alter table for adding credit card column



Check the masked credit card number

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SELECT * FROM employees;

REVERT;
```

Results | Messages

View: Table | Chart | Export results

Search

id	first_name	last_name	salary	phone	email	credit_card
3	Karan	Mehta	1955	XXXXXX6655	kXXX@XXXX.co...	XXXX-XXXX-XXXX-4312
2	Riya	Patel	1306	XXXXXX6780	rXXX@XXXX.com	XXXX-XXXX-XXXX-5678
1	Aakash	Sharma	1497	XXXXXX3210	aXXX@XXXX.co...	XXXX-XXXX-XXXX-1234